

Decision Tree

Extra Class



Nguyen-Thuan Duong – TA

Outline

- Introduction
- Rule-based
- Decision Tree
- Classification
- Question

Introduction

Introduction

❖ Getting Started

The screenshot illustrates the process of accessing contests on the AI Vietnam platform, divided into three steps:

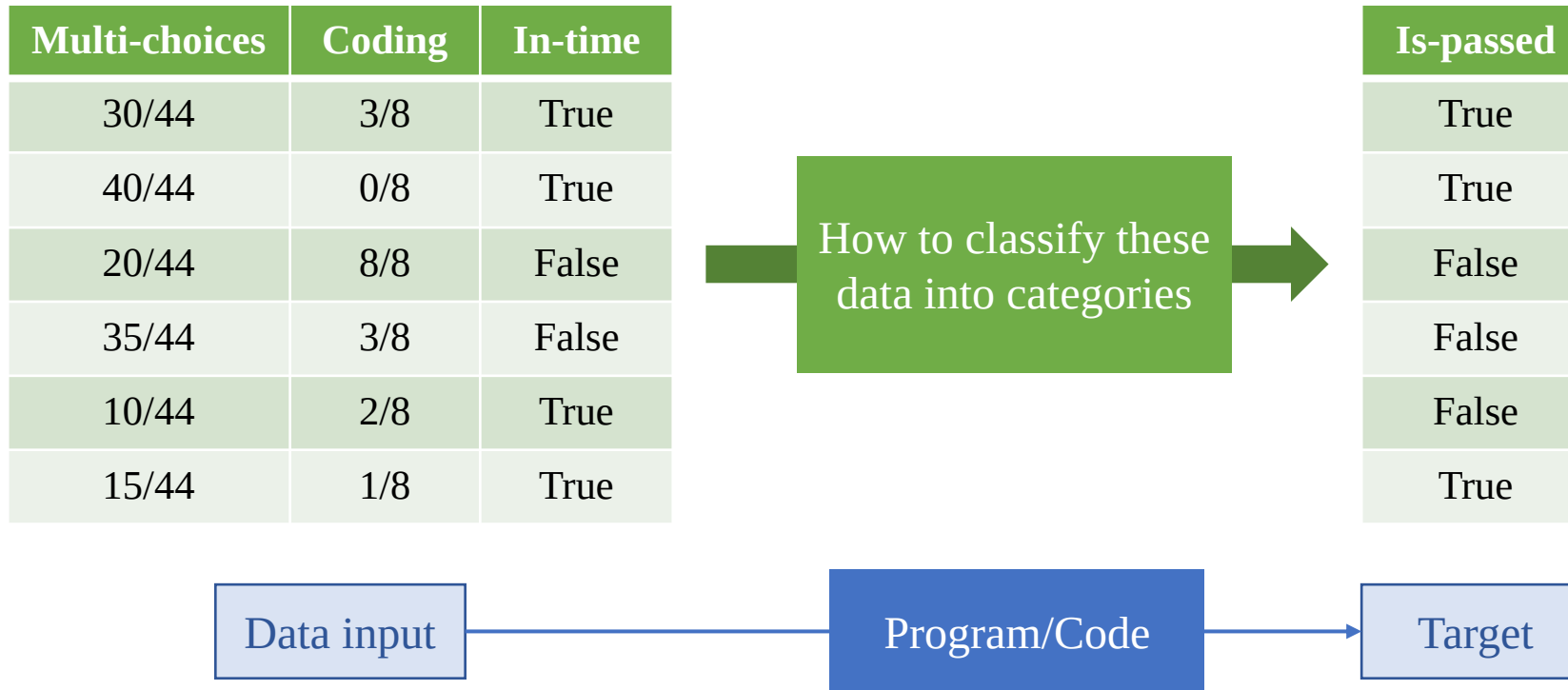
- Step 1:** The user is on the 'Available Contests' page. A red box labeled 'step 1-a' highlights the URL 'aivn-codeless.vercel.app/contest' in the browser's address bar. Another red box labeled 'step 1-b' highlights the 'Sign In' button in the top right corner.
- Step 2:** This step shows two side-by-side login windows. The left window is a 'Sign in to AI Viet Nam' form with a 'Continue with Google' button highlighted by a red box. The right window is a 'Sign in with Google' prompt for 'clerk.shared.lcl.dev', with the 'Continue' button highlighted by a red box.
- Step 3:** After logging in, the user returns to the 'Available Contests' page. A red box highlights the 'Contest' tab in the top navigation bar. A red arrow points from this box to a specific contest card titled 'AIO2024 Contest' which includes 'AIO Module 2 Examination' and a duration of '720 mins'.



Multi-choices	Coding	In-time	Is-passed
30/44	3/8	True	True
40/44	0/8	True	True
20/44	8/8	False	False
35/44	3/8	False	False
10/44	2/8	True	False
15/44	1/8	True	True

Introduction

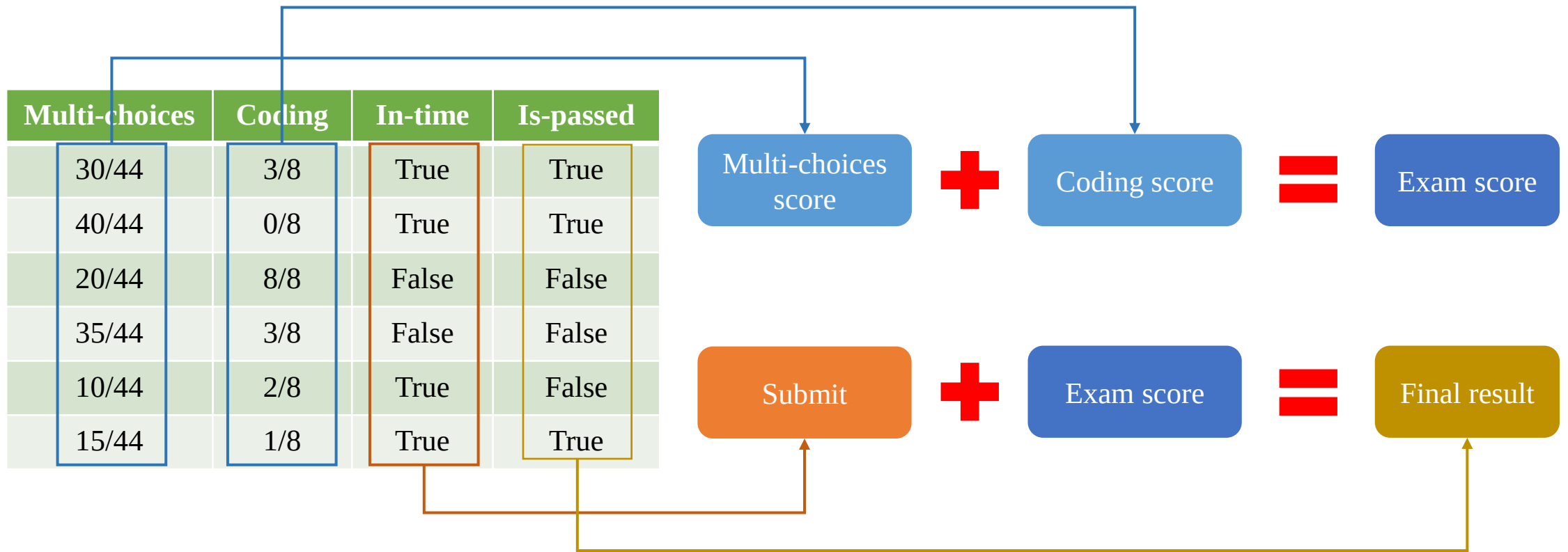
❖ Problem



Rule-based

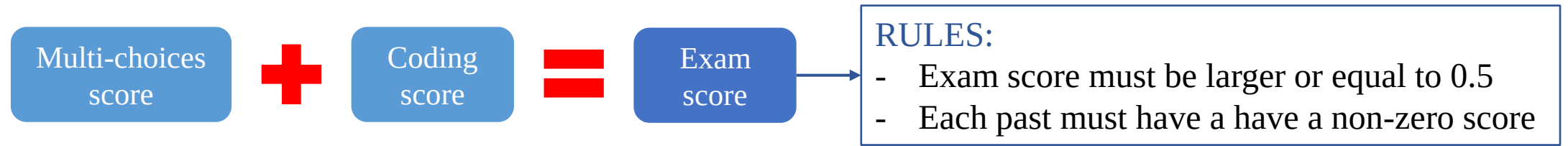
Rule-based

❖ Analysis



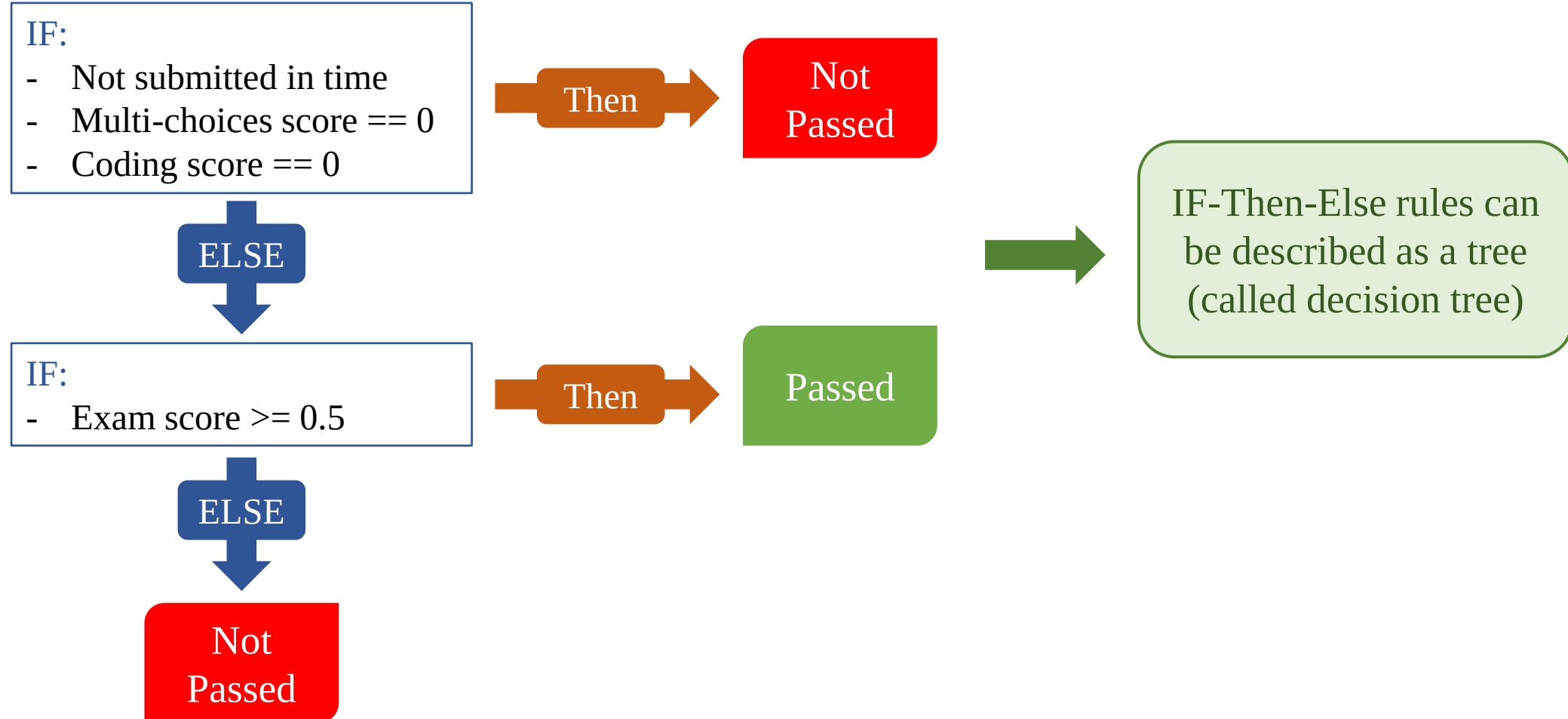
Rule-based

❖ Analysis



Rule-based

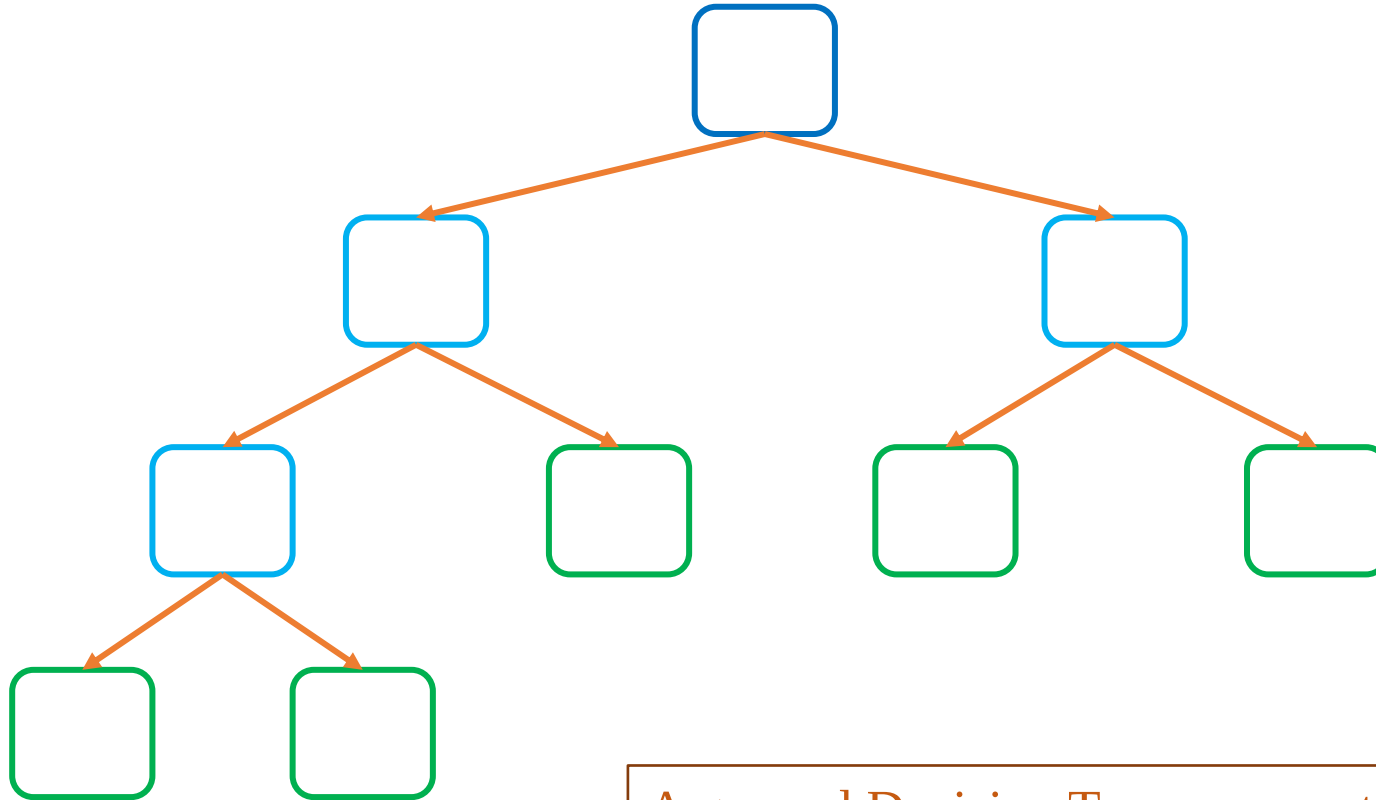
❖ Manual Rules Building



Decision Tree

Decision Tree

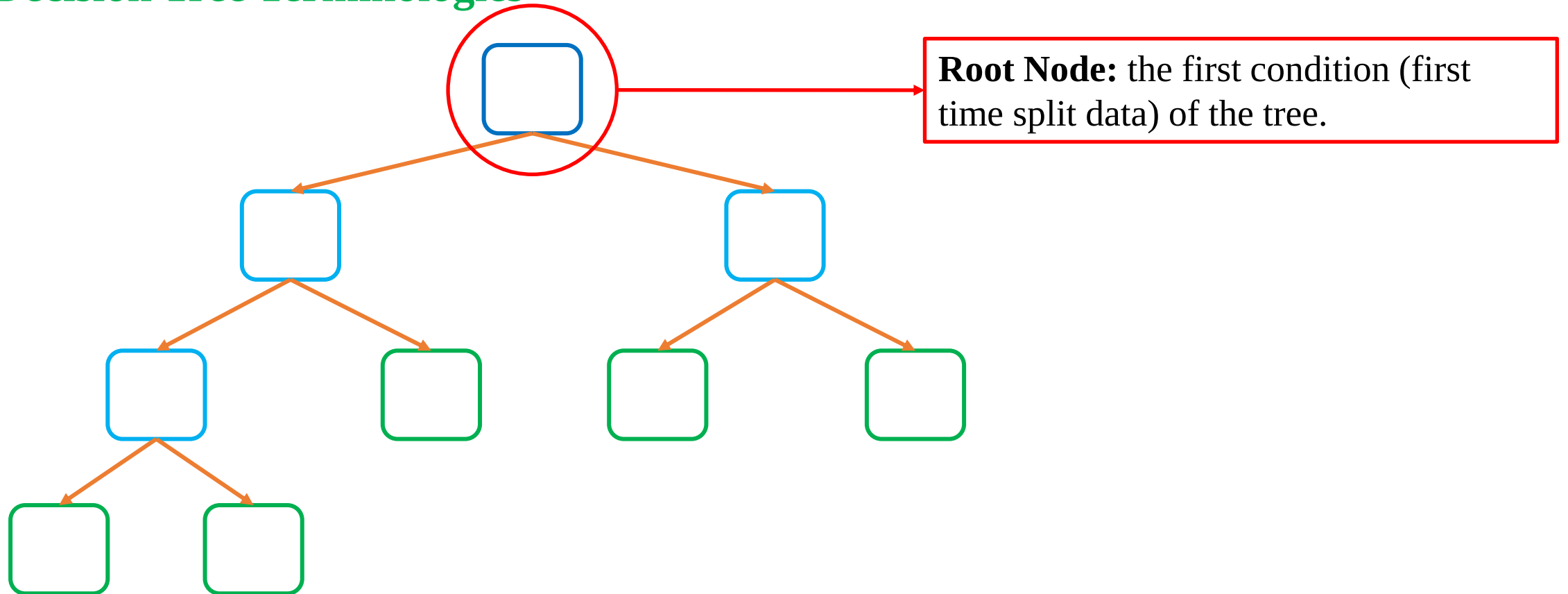
❖ Decision Tree Terminologies



A general Decision Tree may contain many conditions and outcomes.

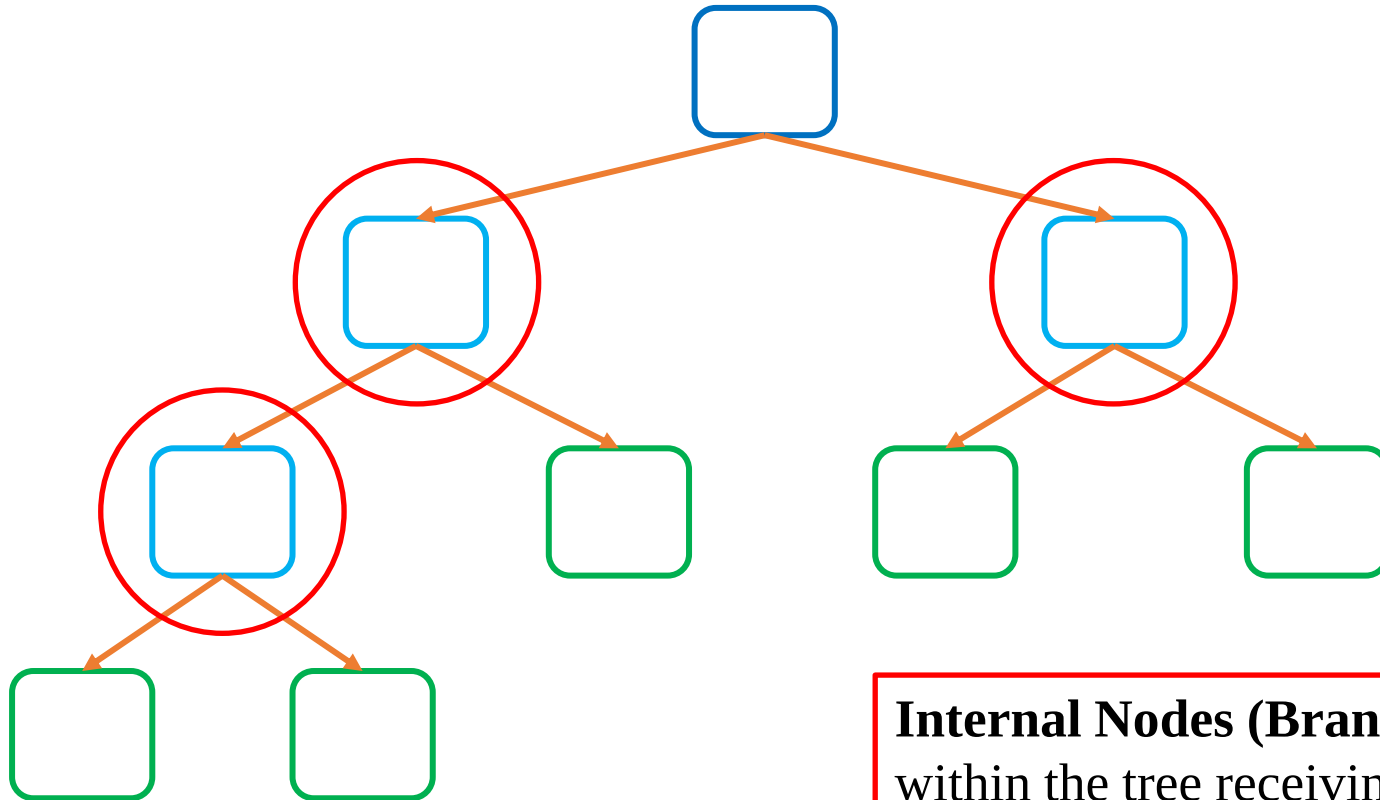
Decision Tree

❖ Decision Tree Terminologies



Decision Tree

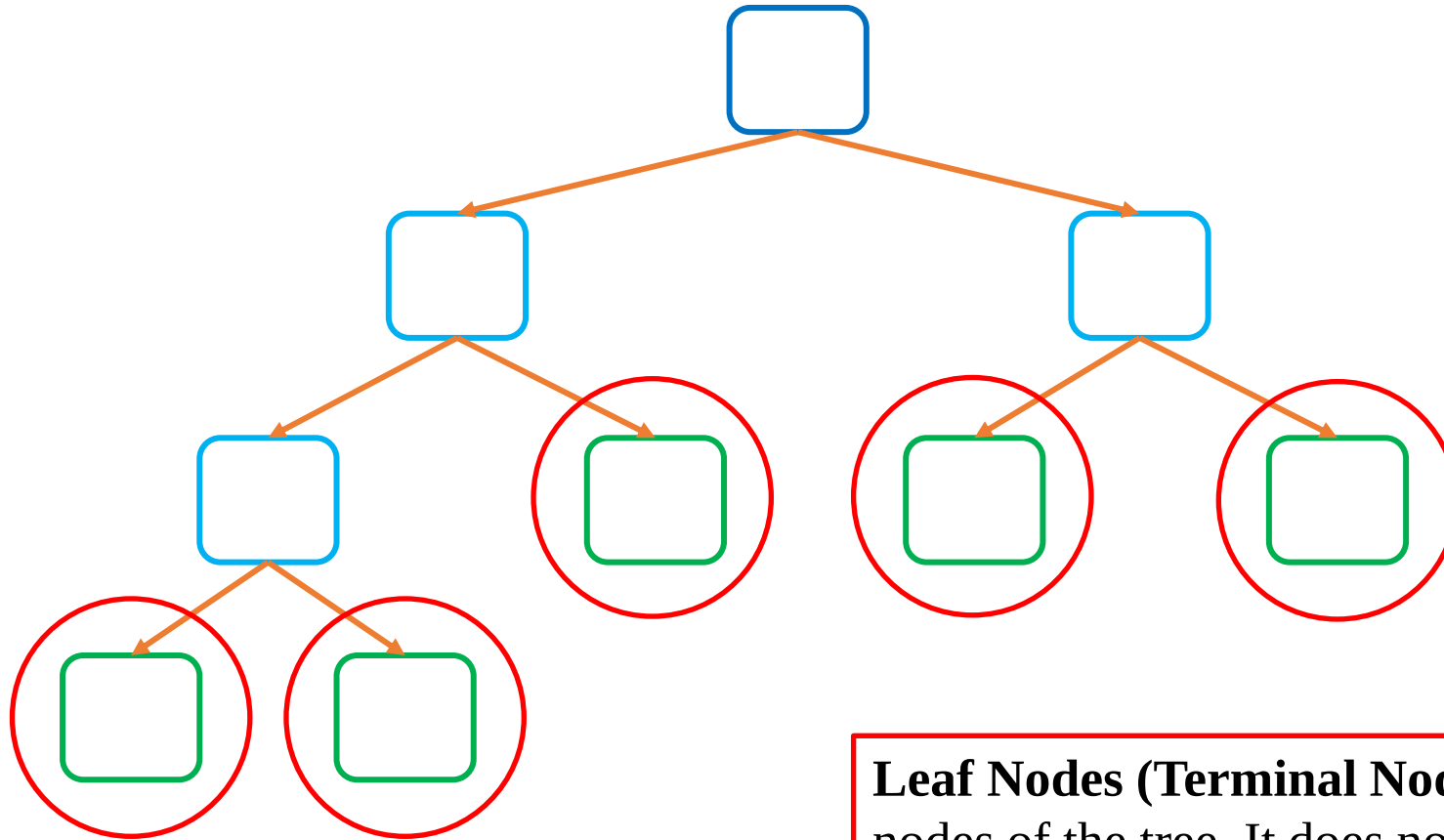
❖ Decision Tree Terminologies



Internal Nodes (Branch Nodes): the conditions within the tree receiving inputs from previous node and process (split) data to new nodes.

Decision Tree

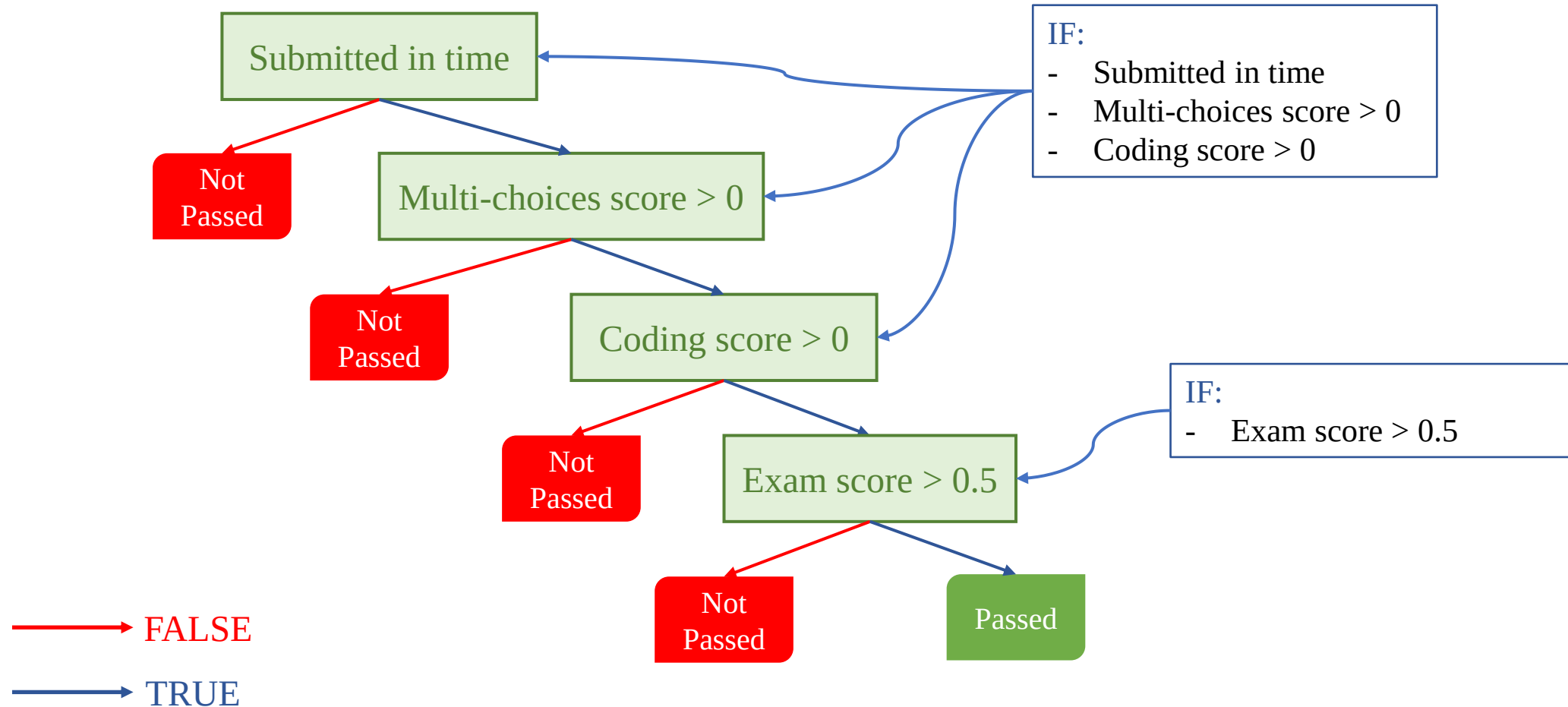
❖ Decision Tree Terminologies



Leaf Nodes (Terminal Nodes): the final decision nodes of the tree. It does not any further splits.

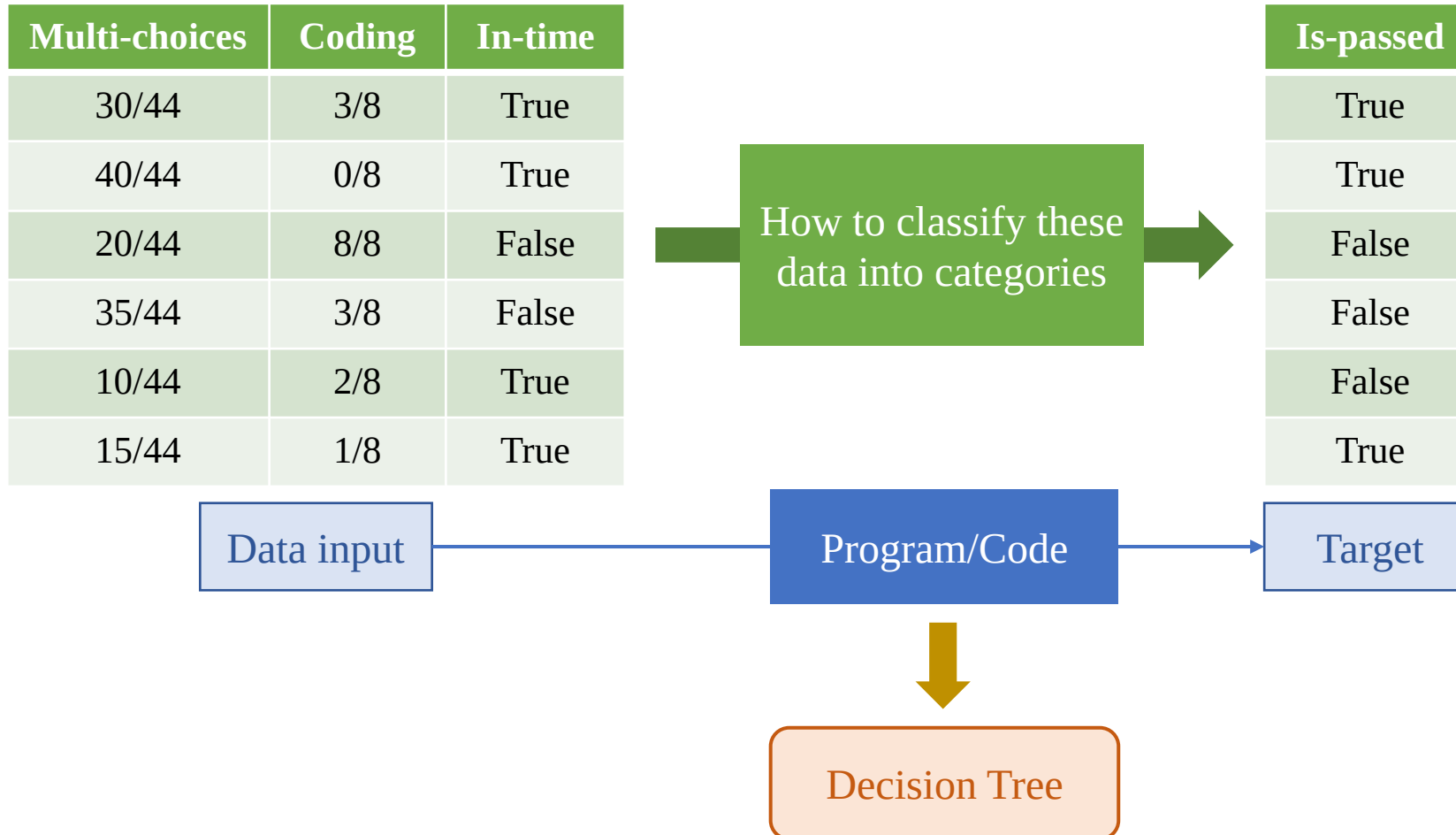
Decision Tree

❖ Manual Tree Building



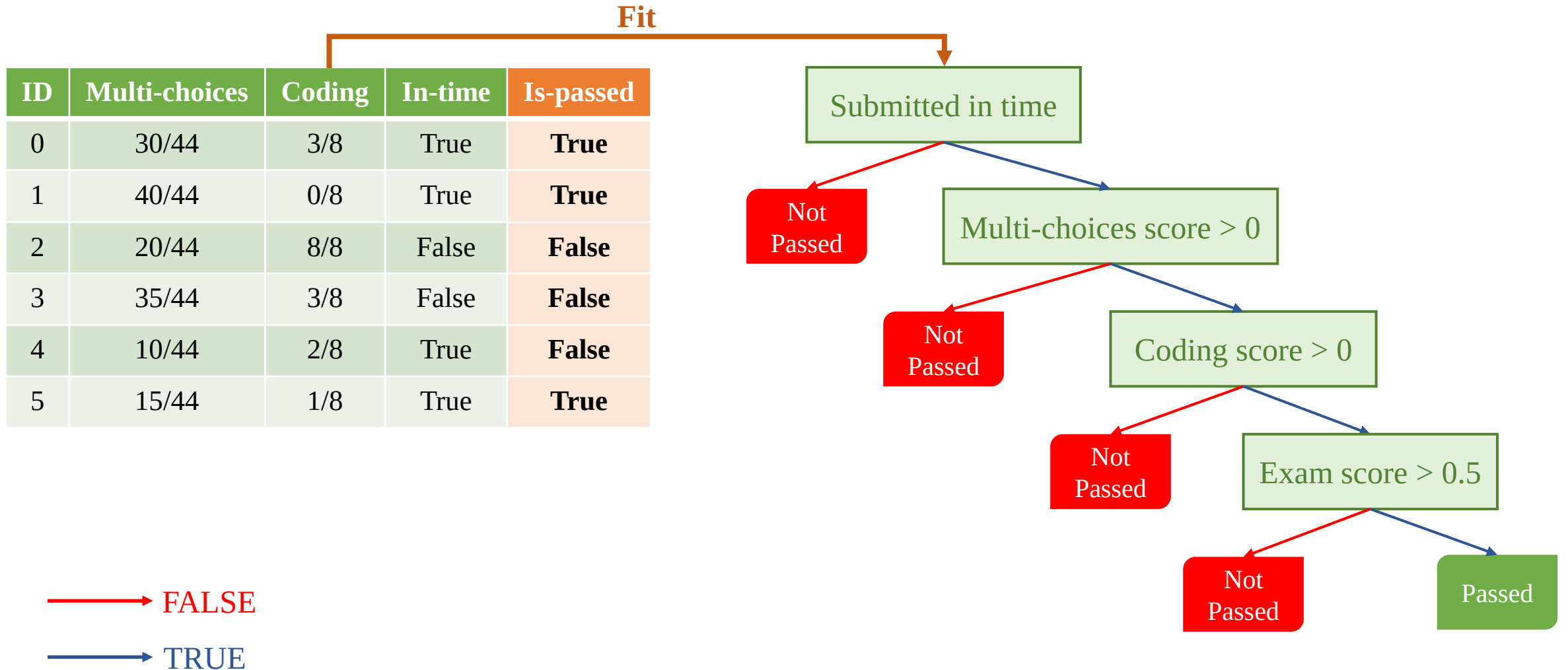
Decision Tree

❖ Problem is handled



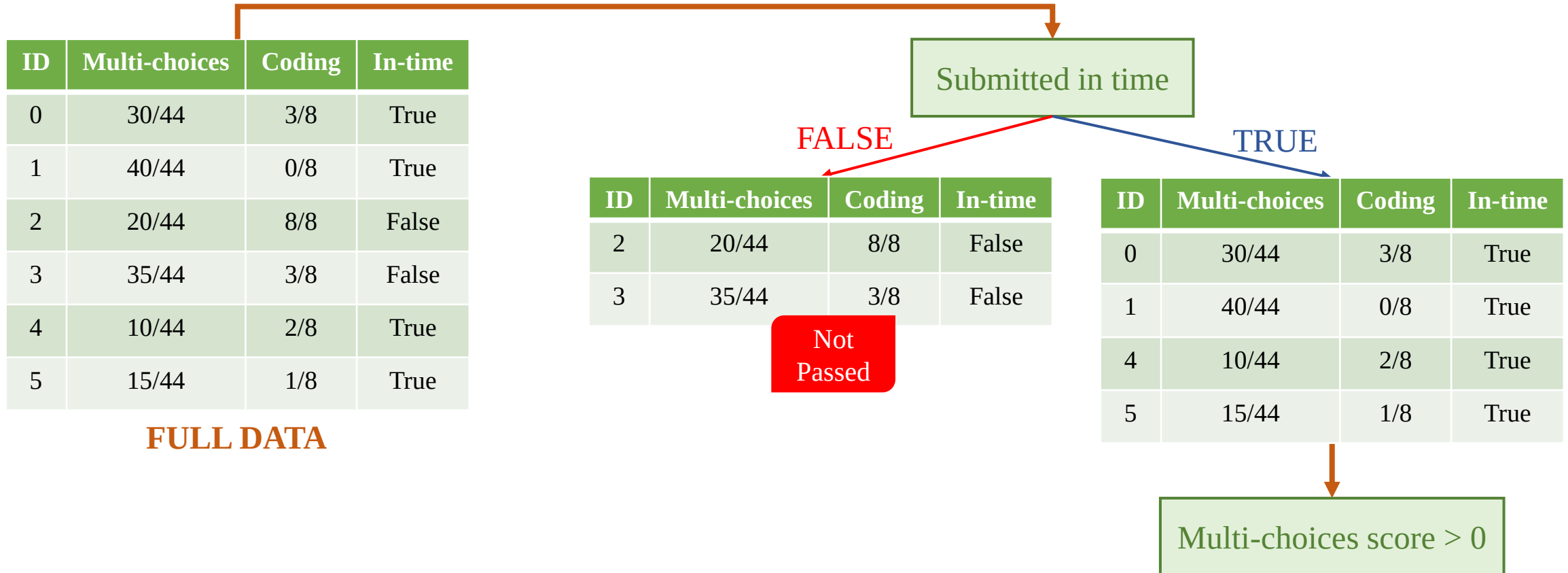
Decision Tree

❖ Fit data to tree



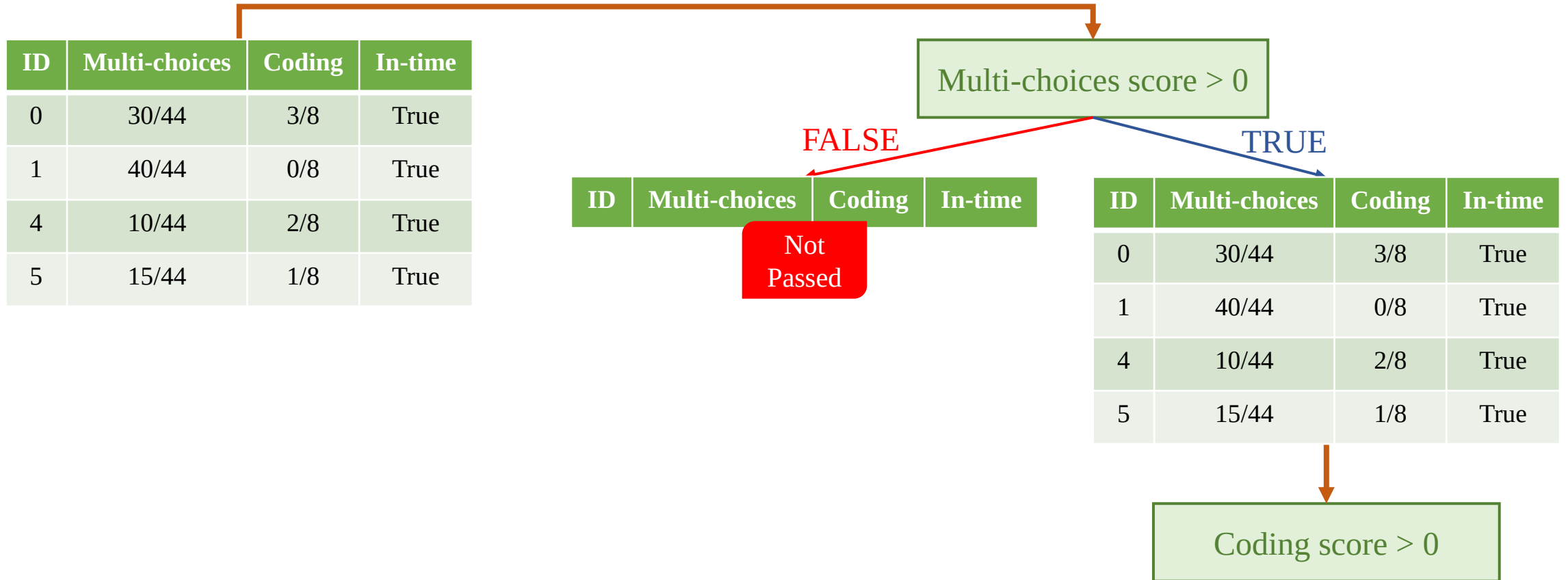
Decision Tree

❖ Fit data to tree



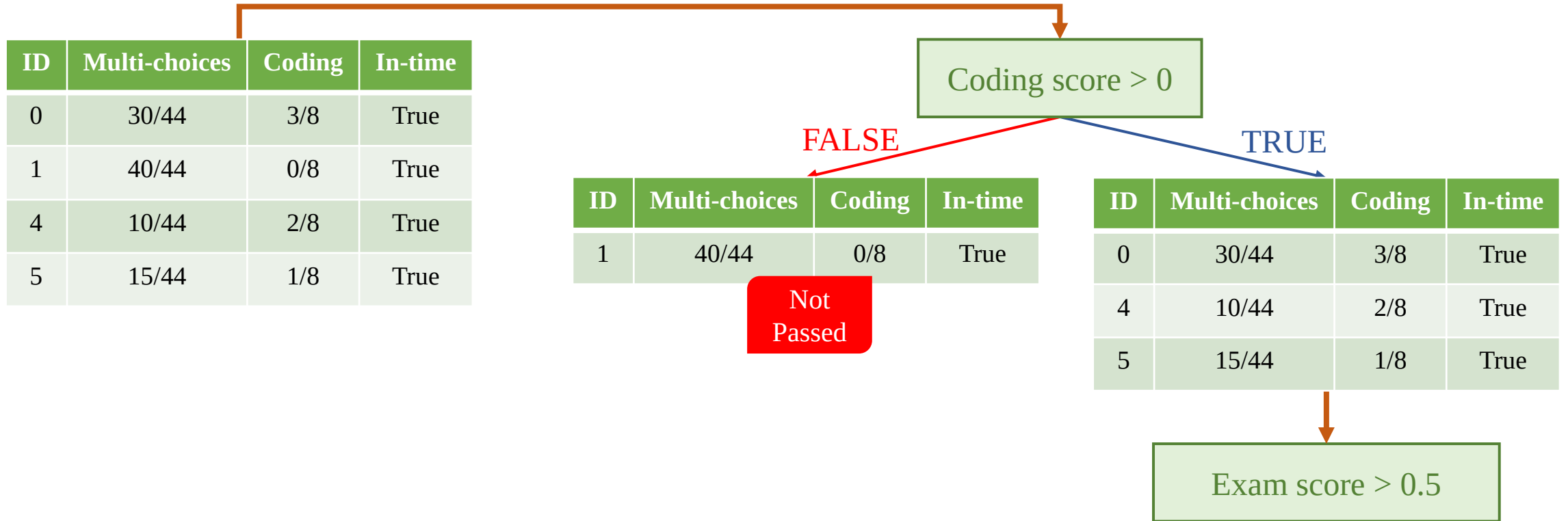
Decision Tree

❖ Fit data to tree



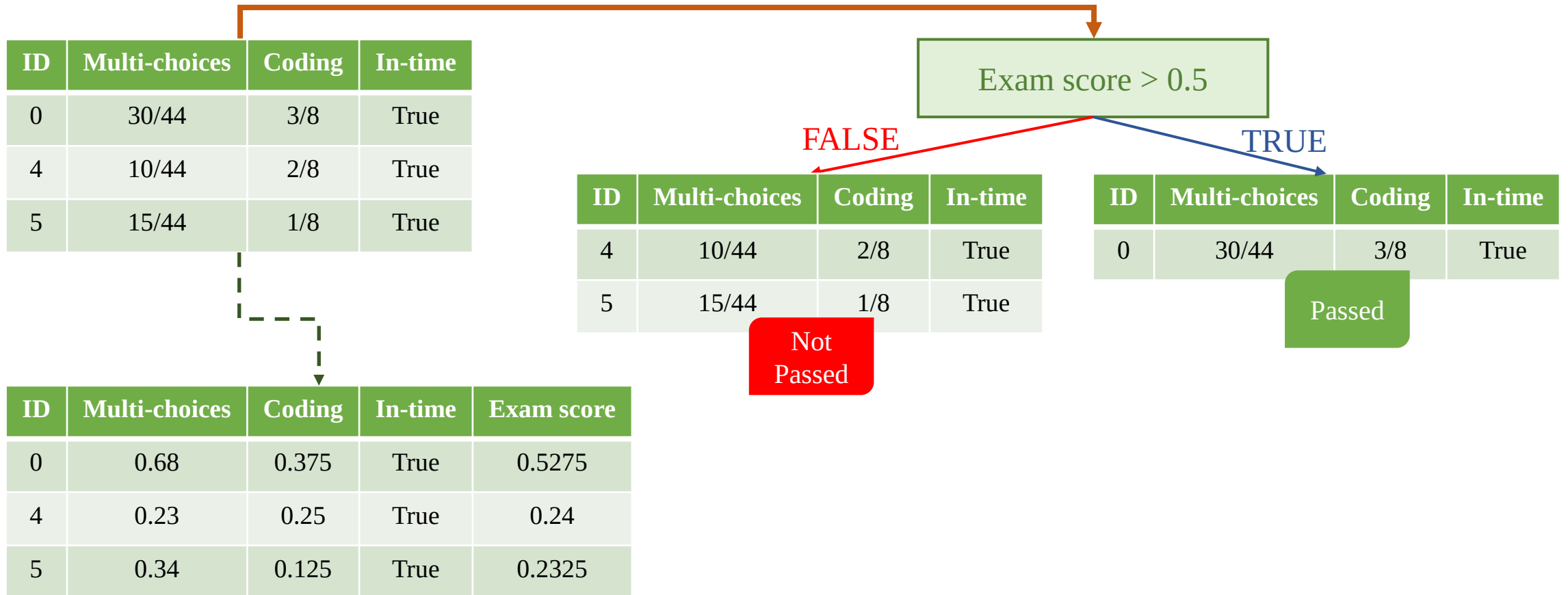
Decision Tree

❖ Fit data to tree



Decision Tree

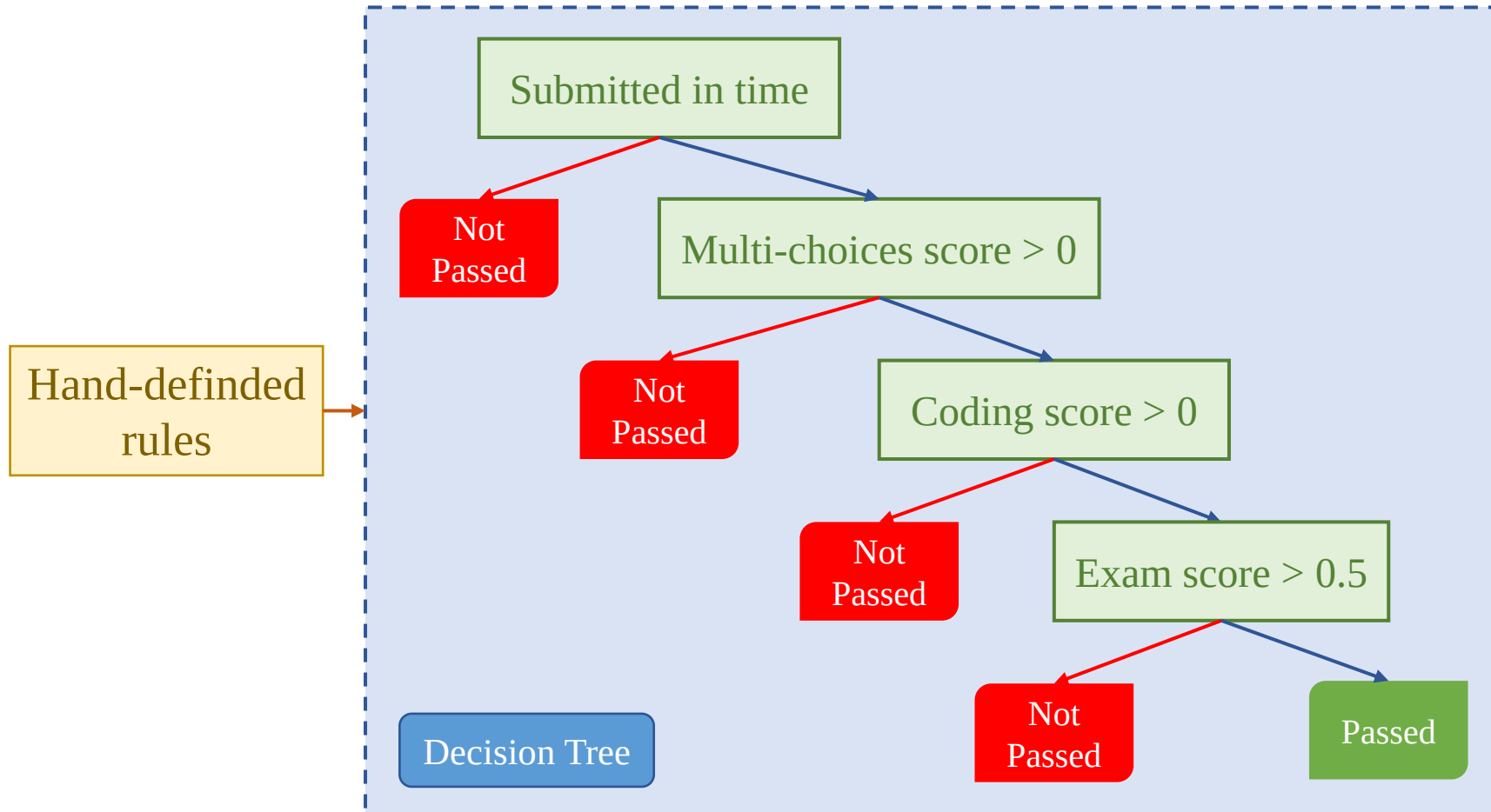
❖ Fit data to tree



Note: weight of Multi-choices and Coding equal to 0.5

Decision Tree

❖ Manual Tree Building



Each rule can build a corresponding tree.

Where is the best rule for dataset to build tree ?

Entropy

Entropy

❖ Mean– Variance – Std Formula

mean

$$E(X) = \sum_{i=1}^N X_i P_X(X_i)$$

variance

$$\begin{aligned} \text{var}(X) &= E\left((X - E(X))^2\right) \\ &= \sum_{i=1}^N (X_i - E(X))^2 P_X(X_i) \end{aligned}$$

Standard
deviation

$$\sigma = \sqrt{\text{var}(X)}$$

Example: $X = \{5, 3, 6, 7, 4\}$

$$\begin{aligned} E(X) &= 5 \times \frac{1}{5} + 3 \times \frac{1}{5} + 6 \times \frac{1}{5} + 7 \times \frac{1}{5} + 4 \times \frac{1}{5} \\ &= 5 \end{aligned}$$

$$\begin{aligned} \text{var}(X) &= \frac{1}{5} [(5 - 5)^2 + (3 - 5)^2 + (6 - 5)^2 + \\ &\quad (7 - 5)^2 + (4 - 5)^2] \\ &= \frac{1}{5} (0 + 4 + 1 + 4 + 1) = 2 \end{aligned}$$

$$\sigma = \sqrt{\text{var}(X)} = 1.41$$

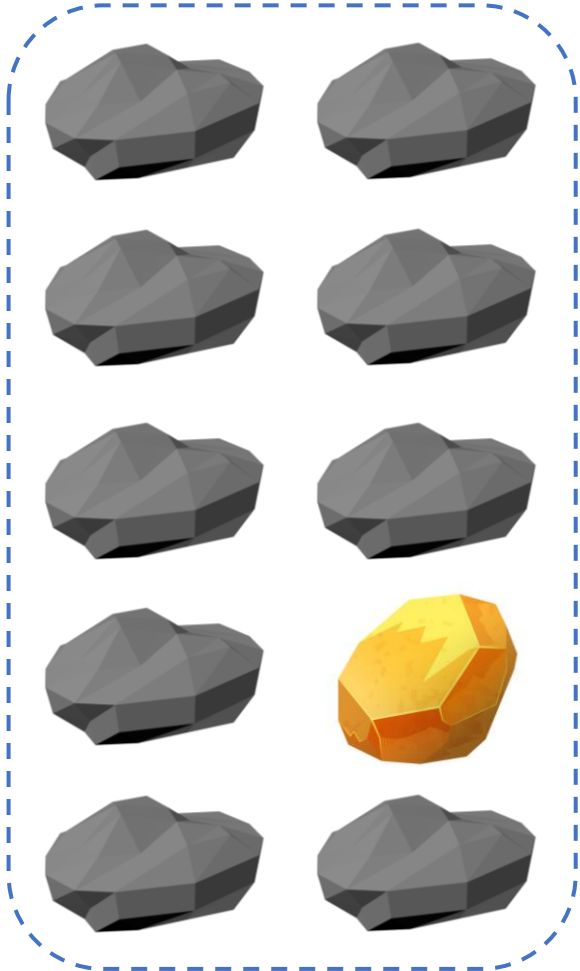
Entropy

❖ Problem



Entropy

❖ Probability



$$P(\text{yellow rock}) = \frac{1}{10} = 0.1$$

$$P(\text{grey rock}) = \frac{9}{10} = 0.9$$



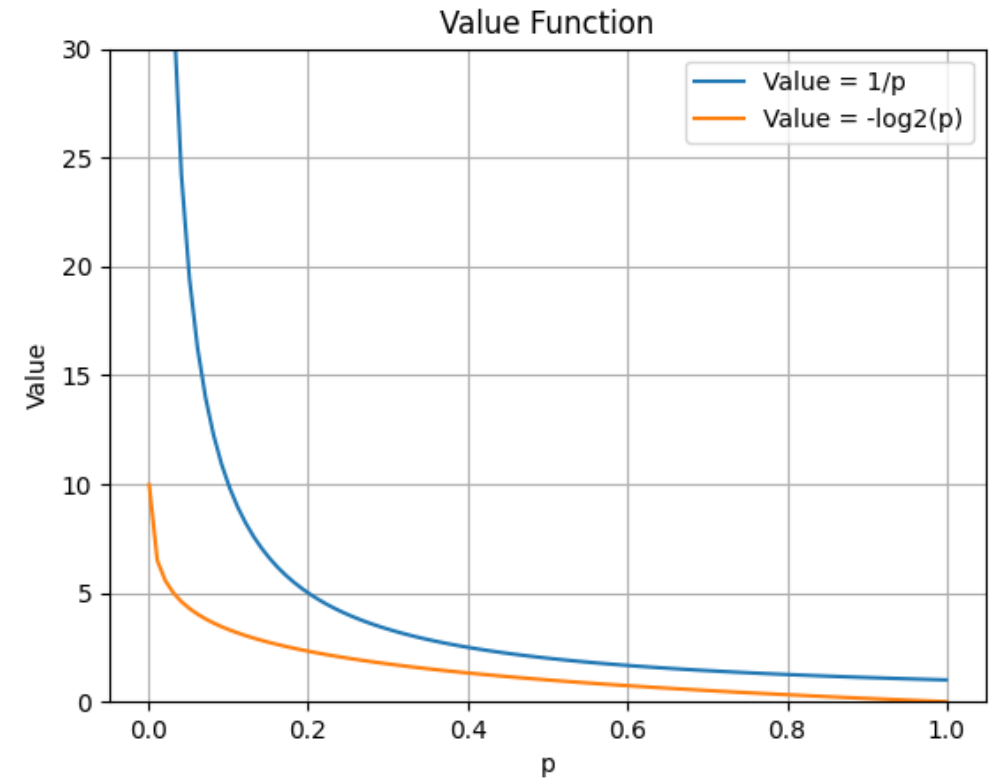
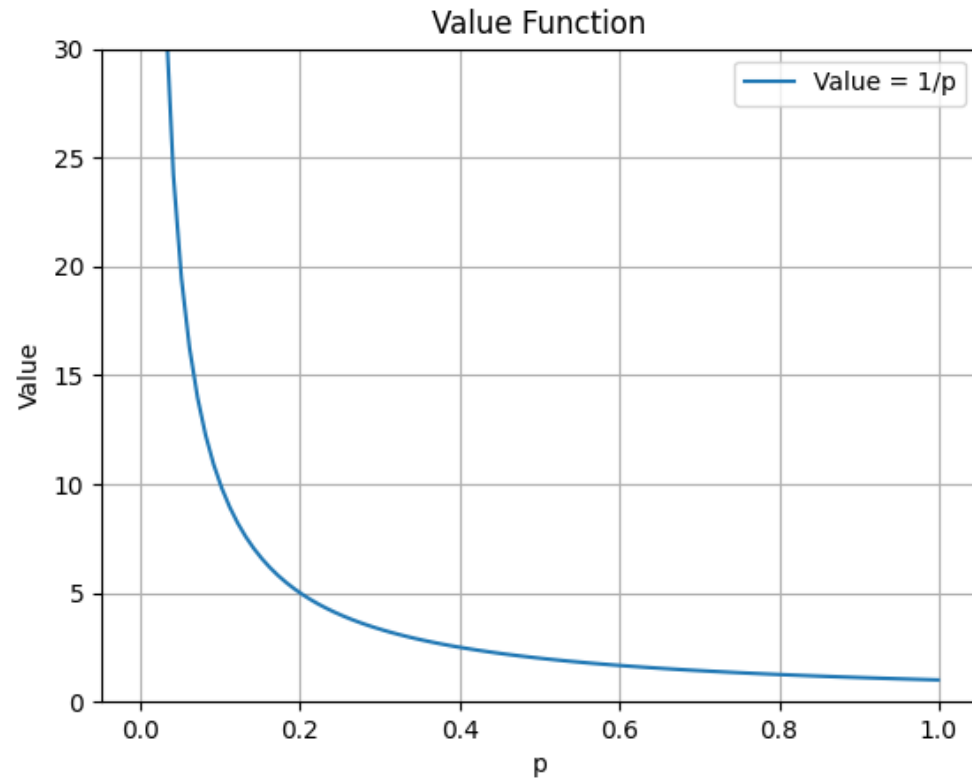
- Low probability
- High value (information)



- Value is scaled with probability

Entropy

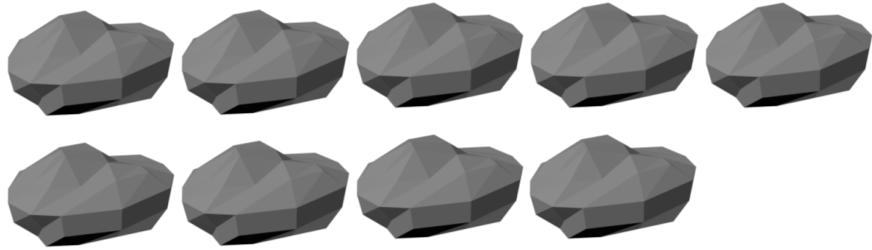
❖ Scale with log2



$$\text{Value}(x) = -\log_2(p(x))$$

Entropy

❖ Mean of Gain



$$0.9 \times -\log_2(0.9) = 0.1368$$

$$0.1 \times -\log_2(0.1) = 0.332$$

Total value:
 $0.1368 + 0.332 = 0.4688$

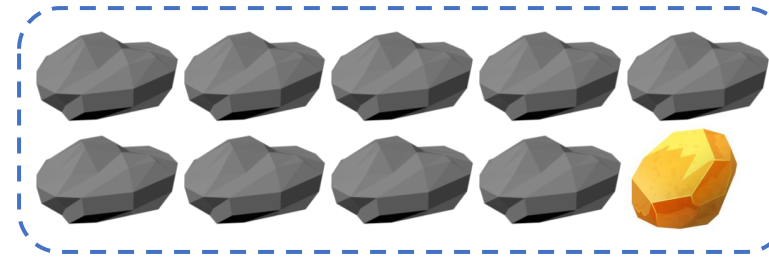
Entropy

Entropy

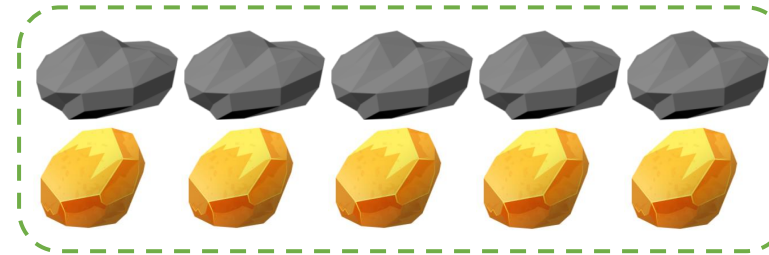
❖ Calculate Entropy

Entropy: Average of information

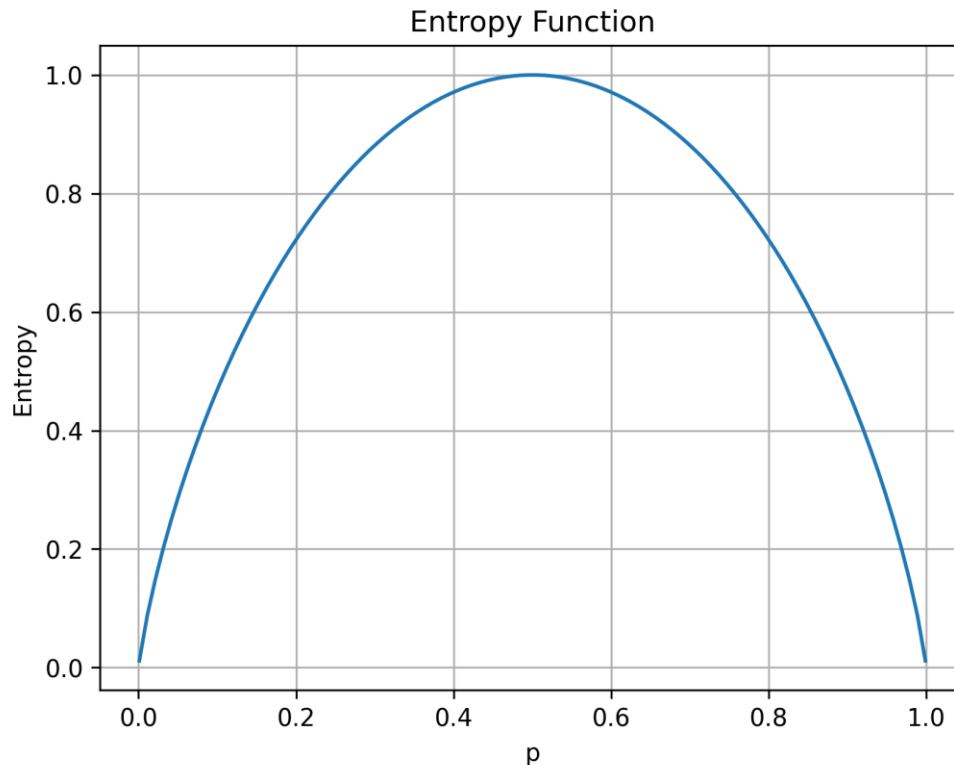
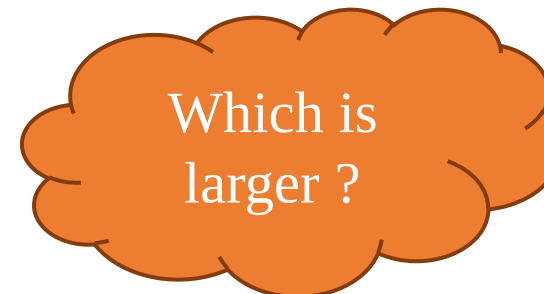
$$H(X) := - \sum_{x \in X} p(x) \log(p(x))$$



Entropy = ?

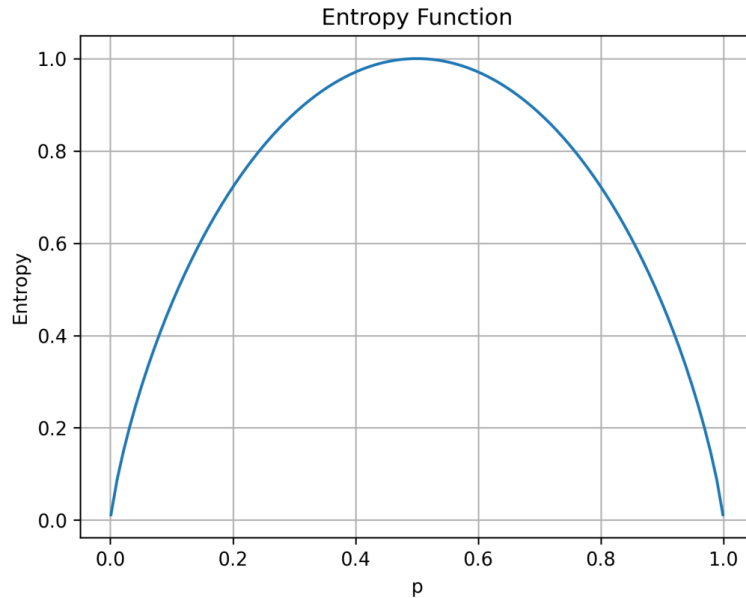


Entropy = ?

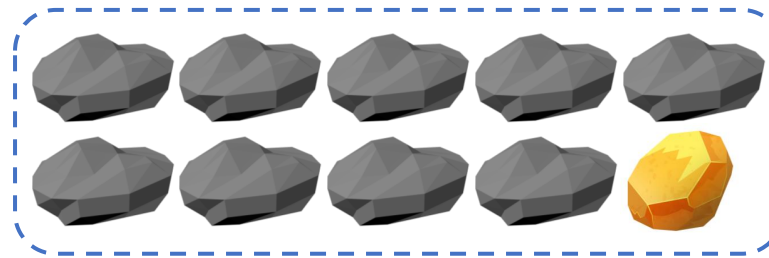


Entropy

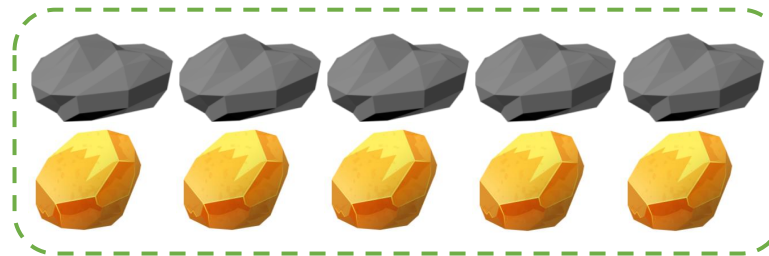
❖ Calculate Entropy



$$H(X) := - \sum_{x \in X} p(x) \log(p(x))$$



$$\begin{aligned} &= -0.9 \log_2(0.9) - 0.1 \log_2(0.1) \\ &= 0.4688 \end{aligned}$$

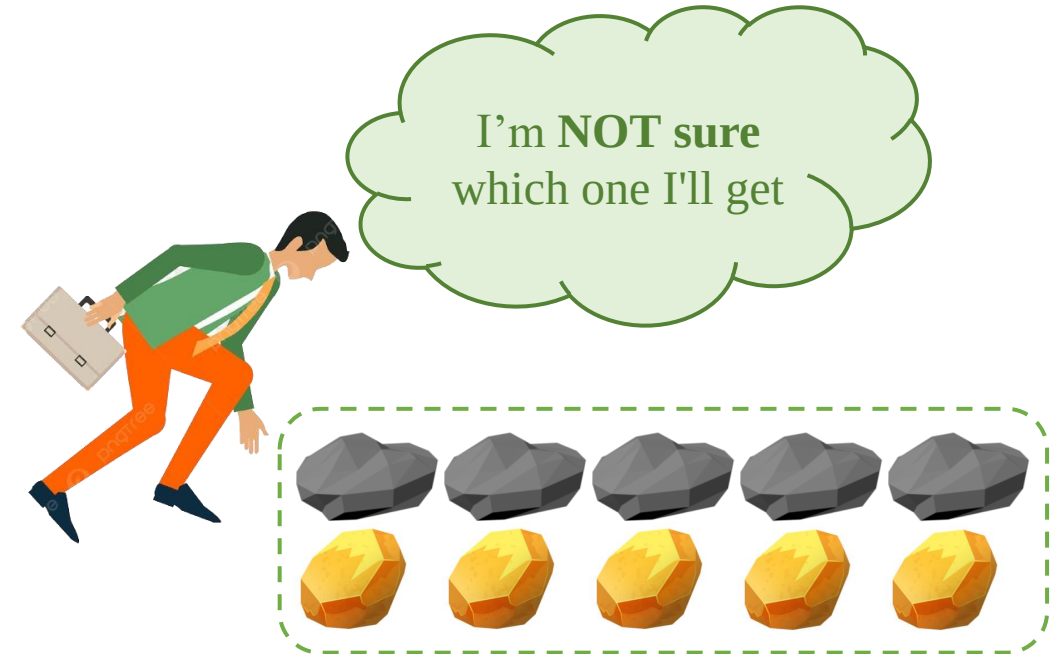
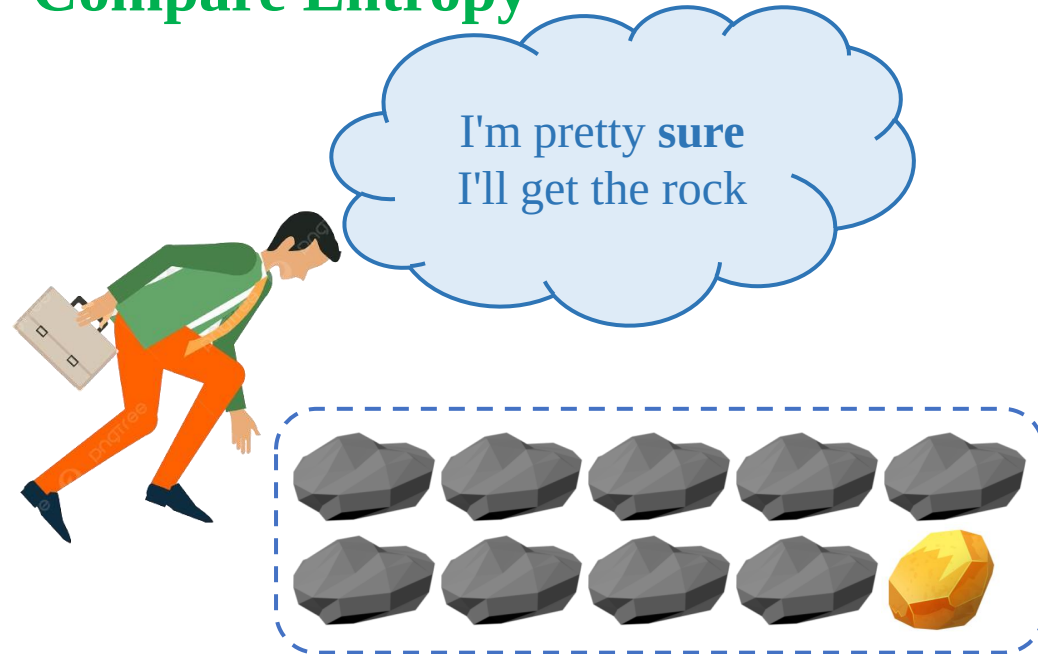


$$\begin{aligned} &= -0.5 \log_2(0.5) - 0.5 \log_2(0.5) \\ &= 1.0 \end{aligned}$$

What exactly
does Entropy
describe?

Entropy

❖ Compare Entropy

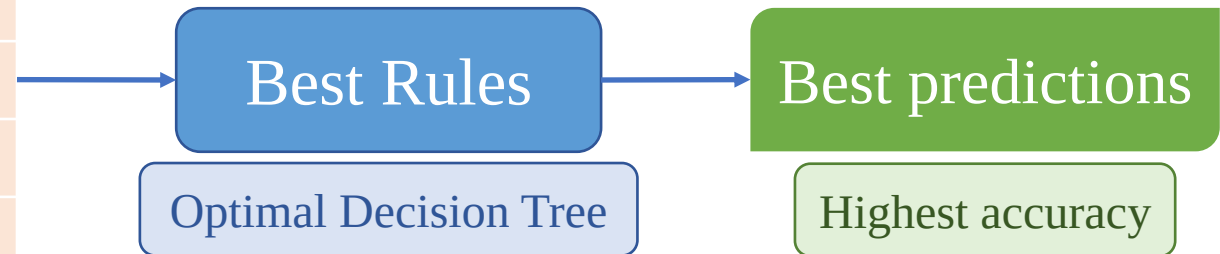


Entropy is used to measure disorder, randomness, or uncertainty.

Classification Tree

❖ Way to find the best rule

ID	Multi-choices	Coding	In-time	Is-passed
0	30/44	3/8	True	True
1	40/44	0/8	True	True
2	20/44	8/8	False	False
3	35/44	3/8	False	False
4	10/44	2/8	True	False
5	15/44	1/8	True	True



Entropy

❖ Measure Information

Multi-choices	Is-passed
30/44	True
40/44	True
20/44	False
35/44	False
10/44	False
15/44	True



Coding	Is-passed
3/8	True
0/8	True
8/8	False
3/8	False
2/8	False
1/8	True



In-time	Is-passed
True	True
True	True
False	False
False	False
True	False
True	True



Entropy

❖ Information Gain (IG)

$$IG = Entropy(parent) - \left[\frac{Average}{Weighted} \right] Entropy(children)$$

$$IG(S, F) = E(S) - \sum_{f \in F} \frac{|S_f|}{S} E(S_f)$$

Classification Tree

Classification Tree

❖ Change dataset

Outlook	Temp	Humidity	Wind	Play Tennis
Sunny	Hot	High	Weak	No
Sunny	Hot	High	Strong	No
Overcast	Hot	High	Weak	Yes
Rain	Mild	High	Weak	Yes
Rain	Cool	Normal	Weak	Yes
Rain	Cool	Normal	Strong	No
Overcast	Cool	Normal	Strong	Yes
Sunny	Mild	High	Weak	No
Sunny	Cool	Normal	Weak	Yes
Rain	Mild	Normal	Weak	Yes
Sunny	Mild	Normal	Strong	Yes
Overcast	Mild	High	Strong	Yes
Overcast	Hot	Normal	Weak	Yes
Rain	Mild	High	Strong	No

Entropy:

$$E(S) = - \sum_{c \in C} p_c \log_2 p_c$$

Information Gain

$$IG(S, F) = E(S) - \sum_{f \in F} \frac{|S_f|}{|S|} E(S_f)$$

$$S = \{9: Yes, 5: No\} \longrightarrow E(S) = -\frac{9}{14} \log_2 \left(\frac{9}{14} \right) - \frac{5}{14} \log_2 \left(\frac{5}{14} \right) = 0.94$$

Parent Entropy

Classification Tree

❖ Children IG

Outlook	Temp	Humidity	Wind	Play Tennis
Sunny	Hot	High	Weak	No
Sunny	Hot	High	Strong	No
Overcast	Hot	High	Weak	Yes
Rain	Mild	High	Weak	Yes
Rain	Cool	Normal	Weak	Yes
Rain	Cool	Normal	Strong	No
Overcast	Cool	Normal	Strong	Yes
Sunny	Mild	High	Weak	No
Sunny	Cool	Normal	Weak	Yes
Rain	Mild	Normal	Weak	Yes
Sunny	Mild	Normal	Strong	Yes
Overcast	Mild	High	Strong	Yes
Overcast	Hot	Normal	Weak	Yes
Rain	Mild	High	Strong	No

Wind	Play Tennis
Strong	No
Strong	No
Strong	Yes
Strong	Yes
Strong	Yes
Strong	No

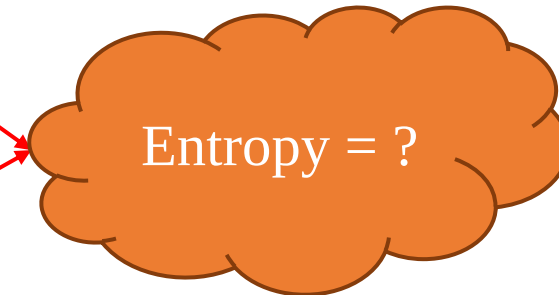
Wind	Play Tennis
Weak	No
Weak	Yes
Weak	Yes
Weak	Yes
Weak	No
Weak	Yes
Weak	Yes
Weak	Yes

Entropy:

$$E(S) = - \sum_{c \in C} p_c \log_2 p_c$$

Information Gain

$$IG(S, F) = E(S) - \sum_{f \in F} \frac{|S_f|}{|S|} E(S_f)$$



Classification Tree

❖ Children IG

Outlook	Temp	Humidity	Wind	Play Tennis
Sunny	Hot	High	Weak	No
Sunny	Hot	High	Strong	No
Overcast	Hot	High	Weak	Yes
Rain	Mild	High	Weak	Yes
Rain	Cool	Normal	Weak	Yes
Rain	Cool	Normal	Strong	No
Overcast	Cool	Normal	Strong	Yes
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Sunny	Cool	Normal	Weak	Yes
Rain	Mild	Normal	Weak	Yes
Sunny	Mild	Normal	Strong	Yes
Overcast	Mild	High	Strong	Yes
Overcast	Hot	Normal	Weak	Yes
Rain	Mild	High	Strong	No

Entropy:

$$E(S) = - \sum_{c \in C} p_c \log_2 p_c$$

Information Gain

$$IG(S, F) = E(S) - \sum_{f \in F} \frac{|S_f|}{|S|} E(S_f)$$

$$S_{weak} = \{6: Yes, 2: No\} \rightarrow E(S_{weak}) = -\frac{6}{8} \log_2 \left(\frac{6}{8}\right) - \frac{2}{8} \log_2 \left(\frac{2}{8}\right) = 0.811$$

$$S_{strong} = \{3: Yes, 3: No\} \rightarrow E(S_{strong}) = -\frac{3}{6} \log_2 \left(\frac{3}{6}\right) - \frac{3}{6} \log_2 \left(\frac{3}{6}\right) = 1$$

$$\begin{aligned} \rightarrow Gain(S, Wind) &= E(S) - \frac{8}{14} E(S_{weak}) - \frac{6}{14} E(S_{strong}) \\ &= 0.94 - \frac{8}{14} * 0.811 - \frac{6}{14} * 1 = 0.048 \end{aligned}$$

Category = 2

Classification Tree

❖ Children IG

Outlook	Temp	Humidity	Wind	Play Tennis
Sunny	Hot	High	Weak	No
Sunny	Hot	High	Strong	No
Overcast	Hot	High	Weak	Yes
Rain	Mild	High	Weak	Yes
Rain	Cool	Normal	Weak	Yes
Rain	Cool	Normal	Strong	No
Overcast	Cool	Normal	Strong	Yes
Sunny	Mild	High	Weak	No
Sunny	Cool	Normal	Weak	Yes
Rain	Mild	Normal	Weak	Yes
Sunny	Mild	Normal	Strong	Yes
Overcast	Mild	High	Strong	Yes
Overcast	Hot	Normal	Weak	Yes
Rain	Mild	High	Strong	No

Entropy:

$$E(S) = - \sum_{c \in C} p_c \log_2 p_c$$

Information Gain

$$IG(S, F) = E(S) - \sum_{f \in F} \frac{|S_f|}{|S|} E(S_f)$$

Category = 3 > 2 →

Combine →

Option_1: Sunny - (Overcast, Rain)
Option_2: Overcast - (Sunny, Rain)
Option_3: Rain - (Sunny, Overcast)

$$S_{Sunny} = \{2: Yes, 3: No\} \rightarrow E(S_{Sunny}) = 0.97$$

$$S_{Overcast, Rain} = \{7: Yes, 2: No\} \rightarrow E(S_{Overcast, Rain}) = 0.764$$

$$IG(S, Option_1)$$

$$= E(S) - \frac{5}{14} E(S_{Sunny}) - \frac{9}{14} E(S_{Overcast, Rain})$$

$$= 0.94 - \frac{5}{14} * 0.97 - \frac{9}{14} * 0.764 = 0.102$$

Classification Tree

❖ Children IG

Outlook	Temp	Humidity	Wind	Play Tennis
Sunny	Hot	High	Weak	No
Sunny	Hot	High	Strong	No
Overcast	Hot	High	Weak	Yes
Rain	Mild	High	Weak	Yes
Rain	Cool	Normal	Weak	Yes
Rain	Cool	Normal	Strong	No
Overcast	Cool	Normal	Strong	Yes
Sunny	Mild	High	Weak	No
Sunny	Cool	Normal	Weak	Yes
Rain	Mild	Normal	Weak	Yes
Sunny	Mild	Normal	Strong	Yes
Overcast	Mild	High	Strong	Yes
Overcast	Hot	Normal	Weak	Yes
Rain	Mild	High	Strong	No

Entropy:

$$E(S) = - \sum_{c \in C} p_c \log_2 p_c$$

Information Gain

$$IG(S, F) = E(S) - \sum_{f \in F} \frac{|S_f|}{|S|} E(S_f)$$

Category = 3 > 2 →

Combine →

Option_1: Sunny - (Overcast, Rain)
Option_2: Overcast - (Sunny, Rain)
Option_3: Rain - (Sunny, Overcast)

$$S_{Overcast} = \{4: Yes, 0: No\} \rightarrow E(S_{Overcast}) = 0$$

$$S_{Sunny, Rain} = \{5: Yes, 5: No\} \rightarrow E(S_{Sunny, Rain}) = 1$$

$$IG(S, Option_2)$$

$$= E(S) - \frac{4}{14} E(S_{Overcast}) - \frac{10}{14} E(S_{Sunny, Rain})$$

$$= 0.94 - \frac{4}{14} * 0 - \frac{10}{14} * 1 = 0.226$$

$$\text{Gain}(S, \text{Outlook}) = \max \begin{cases} IG(S, Option_1) = 0.102 \\ IG(S, Option_2) = 0.226 \\ IG(S, Option_3) = 0.003 \end{cases}$$

Classification Tree

❖ Children IG

Outlook	Temp	Humidity	Wind	Play Tennis
Sunny	Hot	High	Weak	No
Sunny	Hot	High	Strong	No
Overcast	Hot	High	Weak	Yes
Rain	Mild	High	Weak	Yes
Rain	Cool	Normal	Weak	Yes
Rain	Cool	Normal	Strong	No
Overcast	Cool	Normal	Strong	Yes
Sunny	Mild	High	Weak	No
Sunny	Cool	Normal	Weak	Yes
Rain	Mild	Normal	Weak	Yes
Sunny	Mild	Normal	Strong	Yes
Overcast	Mild	High	Strong	Yes
Overcast	Hot	Normal	Weak	Yes
Rain	Mild	High	Strong	No

Entropy:

$$E(S) = - \sum_{c \in C} p_c \log_2 p_c$$

Information Gain

$$IG(S, F) = E(S) - \sum_{f \in F} \frac{|S_f|}{|S|} E(S_f)$$



Gain(S, Outlook) = 0.226

Gain(S, Temp) = 0.015

Gain(S, Humidity) = 0.151

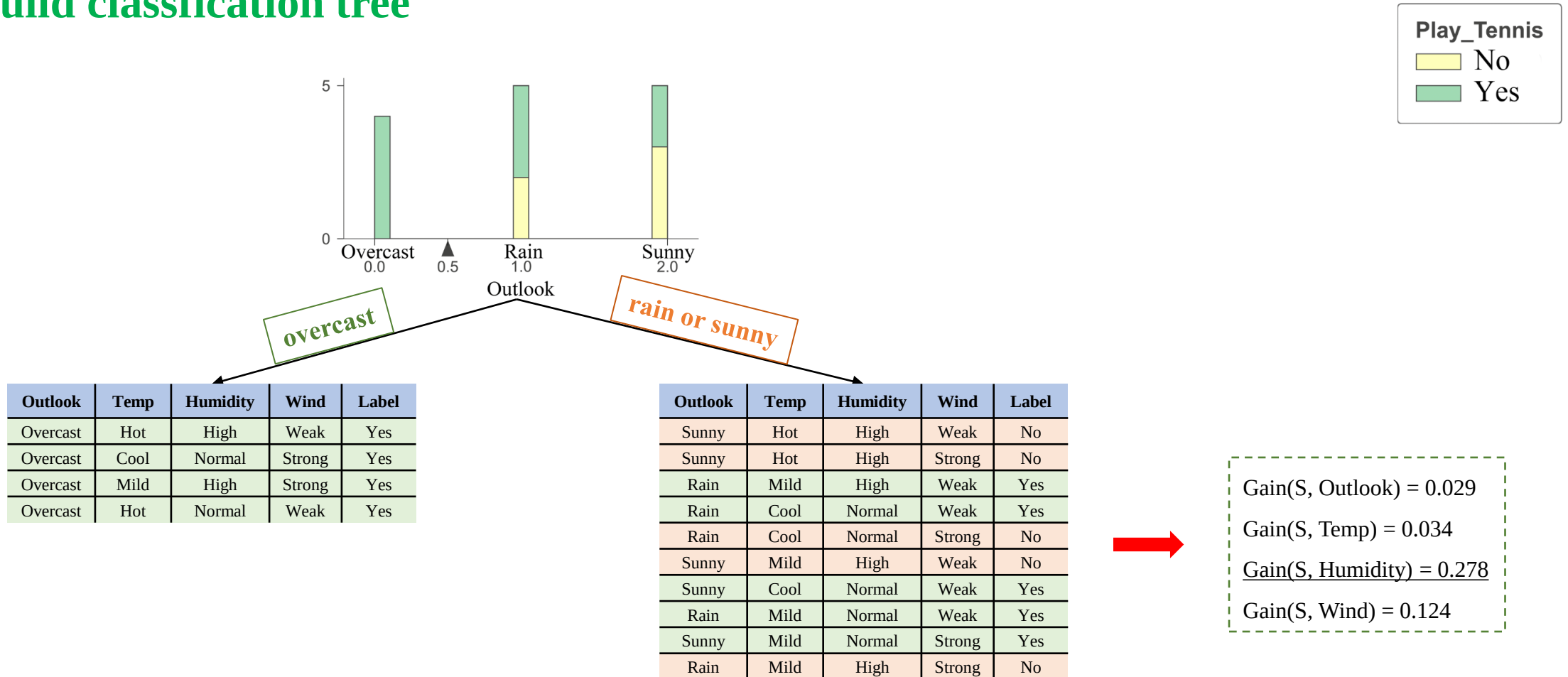
Gain(S, Wind) = 0.048

Choose Outlook
with highest Gain
score for root node

Option_2 is used to
split

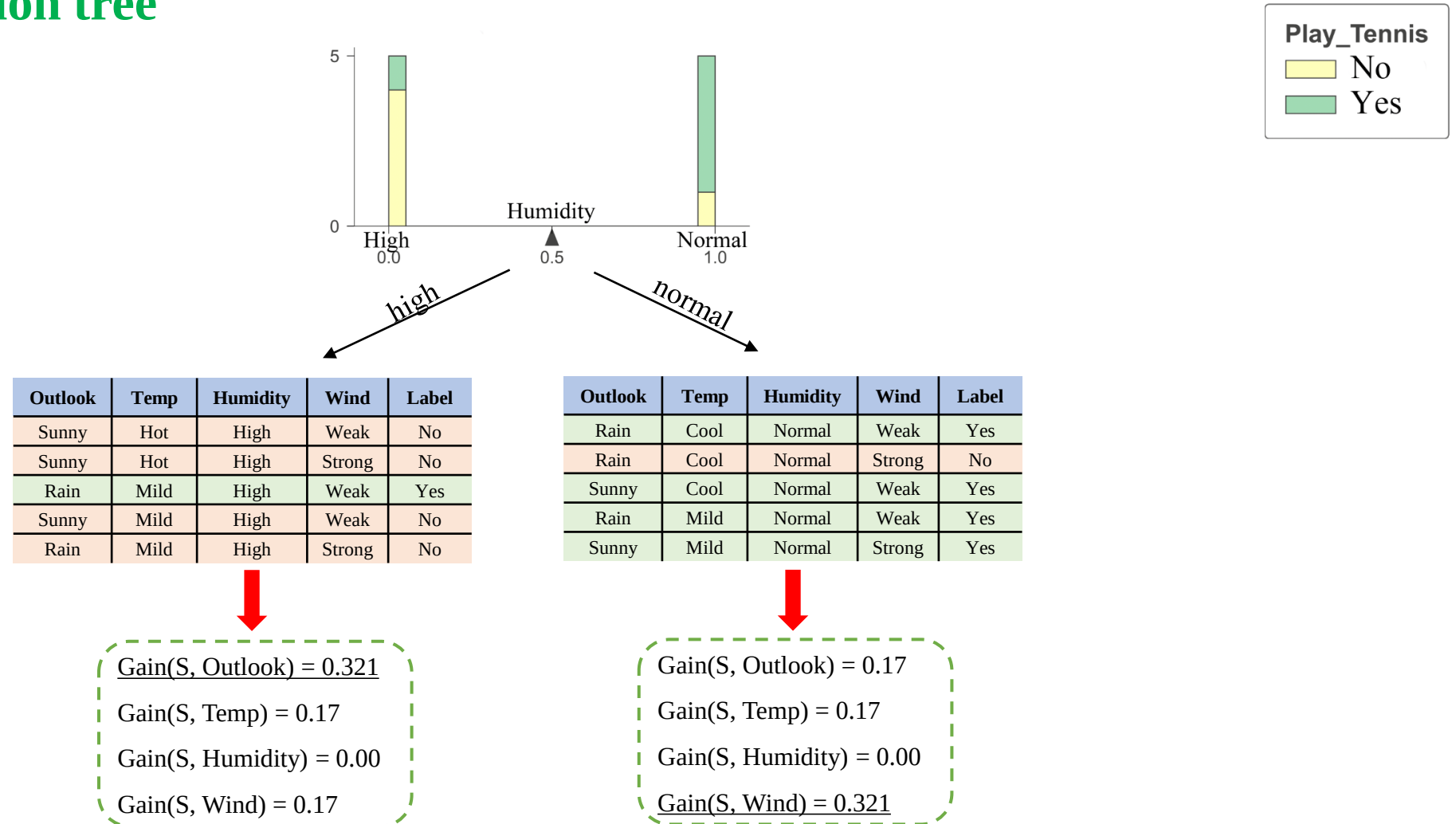
Classification Tree

❖ Build classification tree



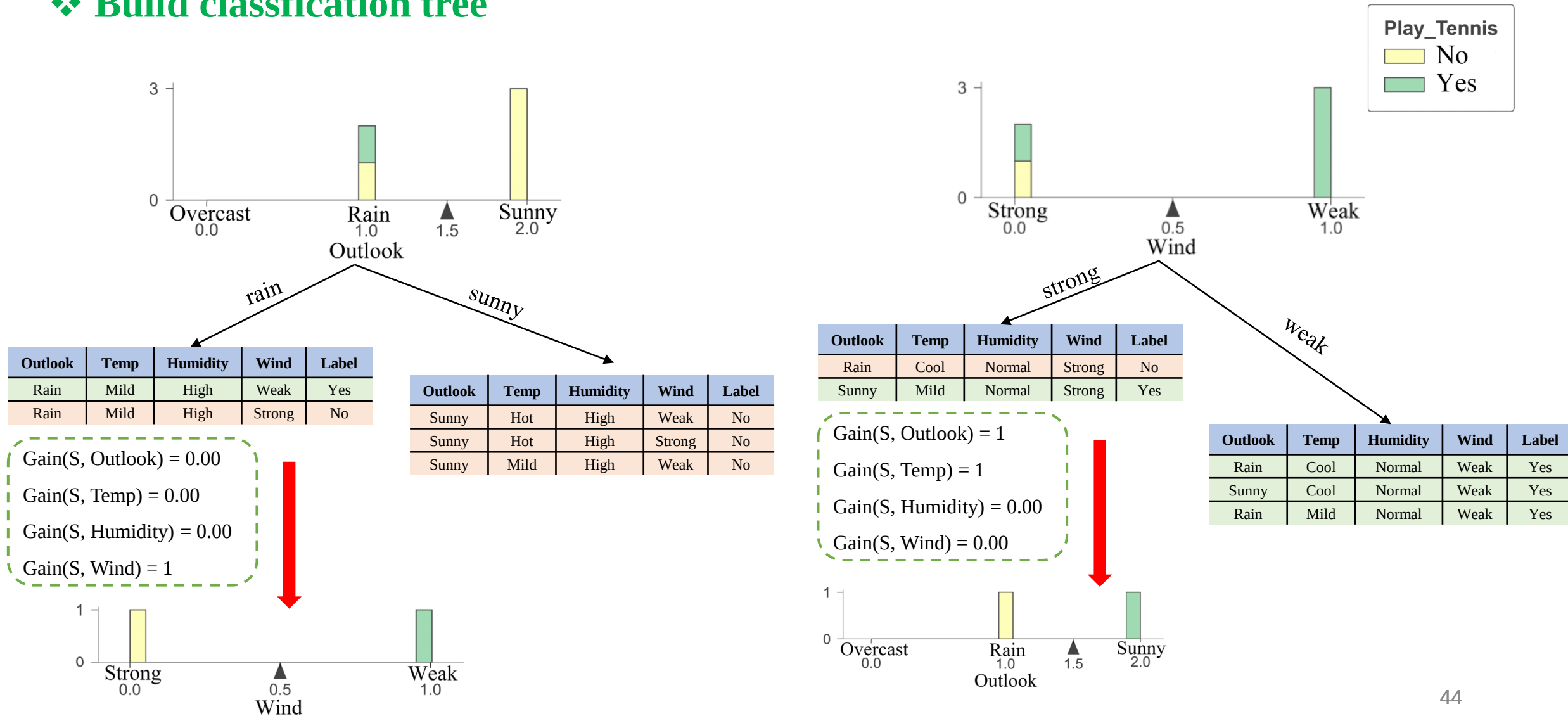
Classification Tree

❖ Build classification tree



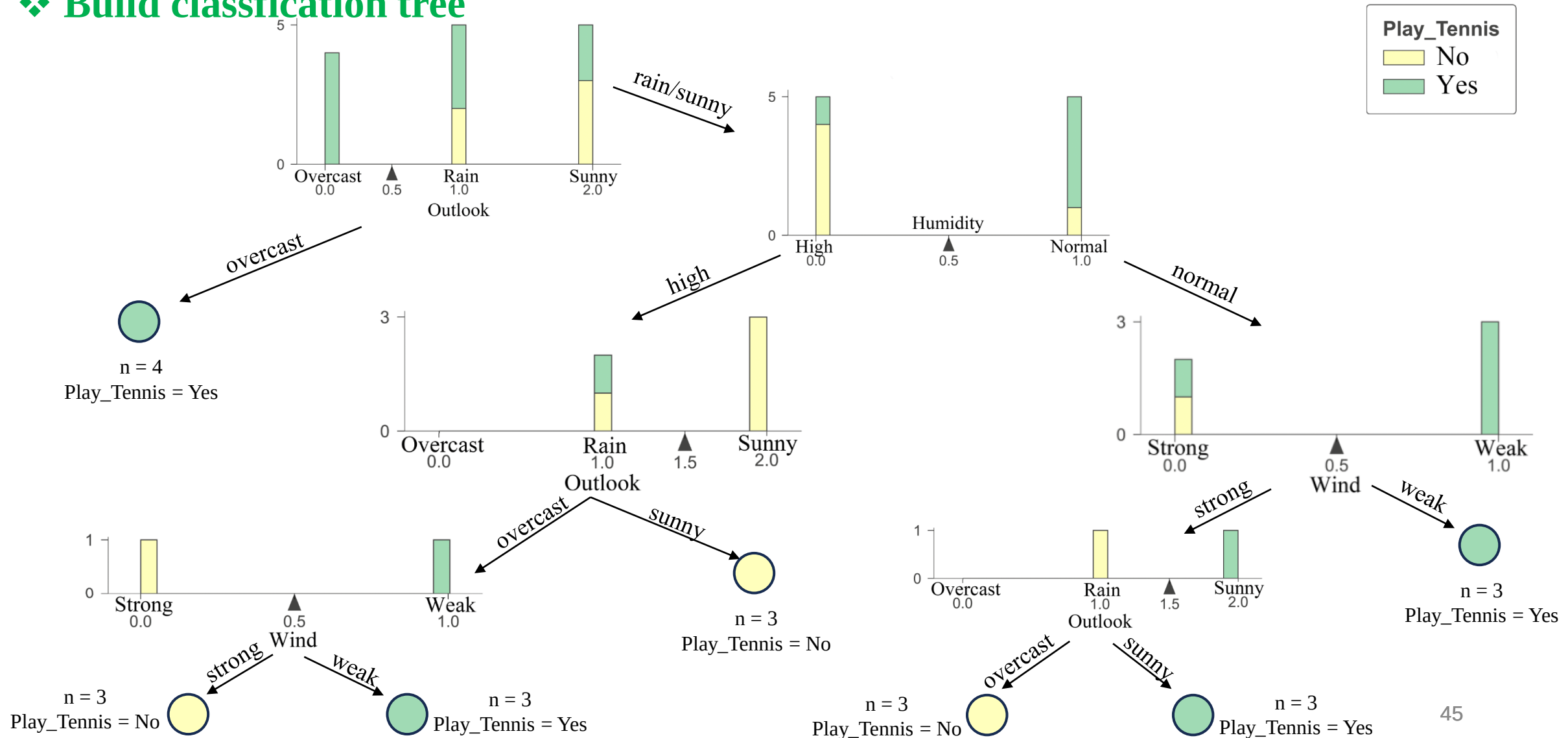
Classification Tree

❖ Build classification tree



Classification Tree

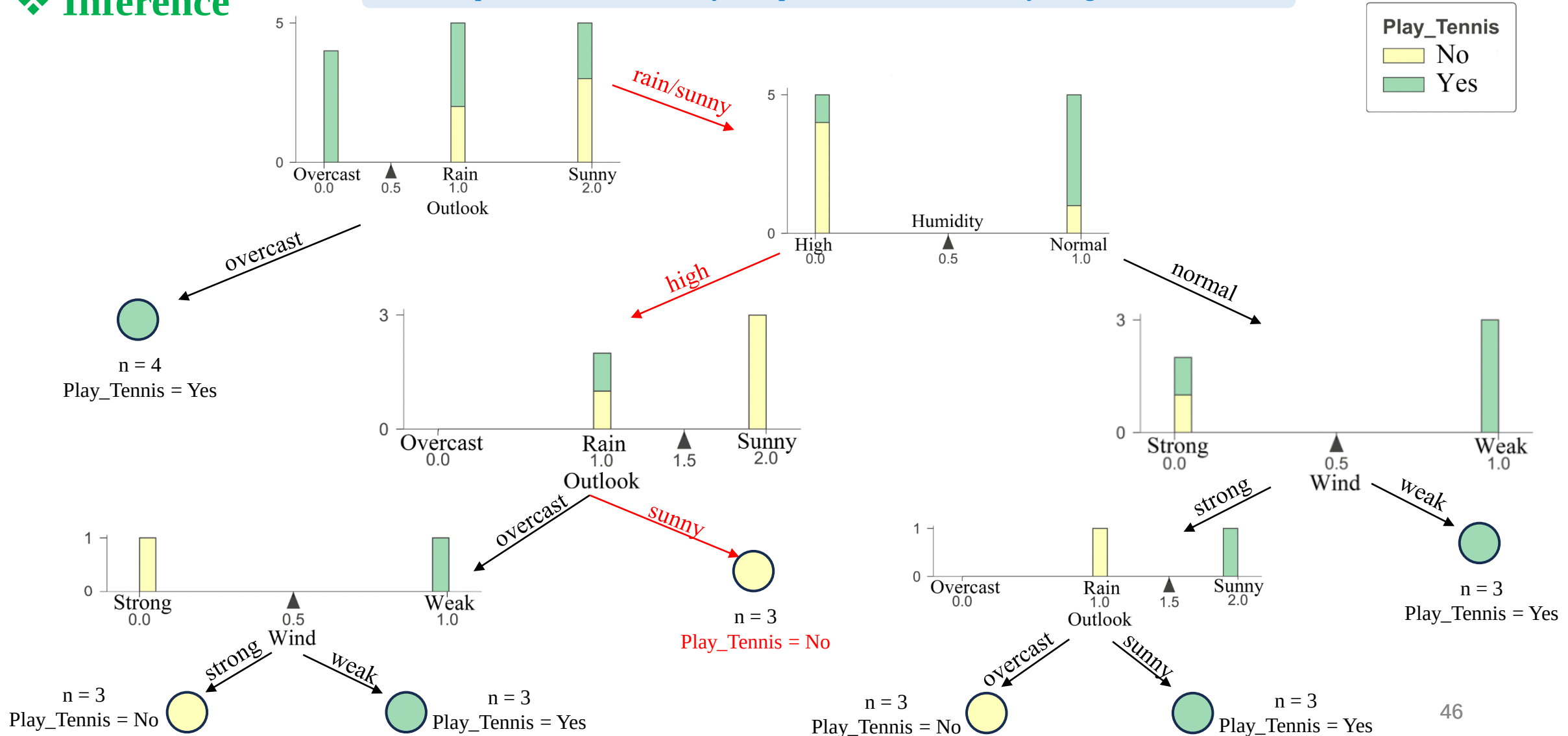
❖ Build classification tree



Classification Tree

❖ Inference

Test sample = <outlook=Sunny, temperature=Hot, humidity=High, Wind=Weak>



Question

